

Holographic Entanglement of Fractal Regions: Complex Dimensions, Non-local Counterterms and Cosmic-Brane Multifractality

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Abstract

We study holographic entanglement entropy of boundary regions whose entangling surface is a self-similar fractal of box dimension D_∂ . Using the Ryu–Takayanagi prescription in Poincaré AdS_{d+1} we show that the radial coordinate acts as a box-counting resolution: in the linearised problem the bulk embedding is the boundary profile convolved with the positive kernel $K_z(y) \propto z^d/(|y|^2 + z^2)^{d-1}$, whose Fourier symbol is $(kz)^{d/2} K_{d/2}(kz)$. This yields the exact identity $-\varepsilon \partial_\varepsilon S_A = (R/4G_N) \text{Vol}(\gamma_A \cap \{z = \varepsilon\})$ and the scaling $-\varepsilon \partial_\varepsilon S_A \asymp N_\varepsilon(\partial A)$. For self-similar boundaries, bulk extensions of the boundary similarities imply a renewal equation whose Mellin poles (complex dimensions) are the zeros of the Moran function; their imaginary parts form the annihilator $\Lambda^\perp$ of the scaling group, and neither real nor imaginary parts are affected by the bulk dynamics, which only enters the residues. Divergences are removed by the scale-non-local counterterm $I_{ct} = -\frac{R}{4G_N} \int_{z_c}^\infty \frac{dz}{z} \Pi_+ \text{Vol}(\gamma_A \cap \{z\})$, and we show why no counterterm of local multiplicative type can do so. We further give a criterion for marginal complex dimensions. Combining Dong’s thermodynamic reading of the refined Rényi entropy with the multifractal formalism, we note that cosmic-brane areas equal the singularity spectrum, $\tilde{S}_q = f(\alpha_q) \ln(1/\varepsilon)$, and that in Einstein gravity its shape is fixed by the bulk. Finally, we propose a spectral fingerprint in lattice fractal-string resonators.

Contents

1	Introduction	2
2	Geometry: Poincaré AdS and the radial resolution	4
3	Fractal area law	4
3.1	Setup	4
3.2	An exact radial identity	5
3.3	The linearised problem and the radial kernel	5
3.4	Content asymptotics for Weierstrass boundaries	6
3.5	Holographic box counting	7
3.6	Numerical test	7
4	Discrete scale invariance and complex dimensions	8
4.1	Self-similar boundaries and the bulk similarity group	8
4.2	Near-boundary decoupling	9
4.3	DSI locking	9
4.4	The non-local counterterm	11
4.5	Numerical illustration	12

5	Marginal complex dimensions	13
5.1	Thread landing and the distance zeta function	13
5.2	The criterion	14
5.3	An explicit lattice example	14
6	Rényi entropies, cosmic branes and multifractality	15
6.1	Dong’s area law and its thermodynamic reading	15
6.2	Cosmic-brane areas as a singularity spectrum	15
6.3	Einstein gravity: the shape of f is fixed by the bulk	16
6.4	Sparse spectra and the $1/c$ tension	17
7	Bit threads and scale transport	17
7.1	Flux depth distribution and geodesic flows	17
7.2	A scale-transport equation	19
8	Experimental fingerprint: lattice fractal-string resonators	20
9	Discussion and limitations	21
A	Numerical methods	23
B	Mellin asymptotics and renewal theory	23

1 Introduction

The Ryu–Takayanagi (RT) formula [1]

$$S_A = \frac{\text{Area}(\gamma_A)}{4G_N} \tag{1}$$

identifies the entanglement entropy of a boundary region A with the area of the minimal bulk surface γ_A homologous to A . Nearly all explicit studies of (1) concern regions with smooth boundaries—balls, strips, and their small deformations—or boundaries with isolated singularities such as corners and cones. In all these cases the ultraviolet structure of S_A is controlled by a *local* expansion: the near-boundary Fefferman–Graham expansion of γ_A [11, 12] determines the divergent terms as integrals of local geometric invariants of ∂A , and holographic renormalisation [13] removes them by local counterterms. The coefficient of the leading area-law divergence is non-universal, while subleading logarithms (in even d , or from corners in $d = 3$) carry universal data of the CFT.

This locality rests on the entangling surface being smooth below some scale. A fractal entangling surface has no such scale: structure is present at every resolution, and ∂A has infinite $(d - 2)$ -volume. Fractal boundaries arise naturally as coarse-grained interfaces, critical domain walls and regions defined on fractal lattices, but the motivation here is structural. In holography the radial coordinate z geometrises the renormalisation group: the dilatation $(x, z) \mapsto (\lambda x, \lambda z)$ is an isometry, and the bulk at depth z encodes the boundary at resolution z . A fractal ∂A is therefore a probe of how the bulk organises an *infinite hierarchy* of scales, and of how the continuous dilatation symmetry of AdS interacts with the *discrete* scale invariance (DSI) of a self-similar set. We will see that this interaction has sharp consequences: complex scaling exponents, log-periodic terms, the breakdown of local counterterms, and a precise dictionary between cosmic branes and multifractal spectra.

Prior work. Our analysis builds on several lines of work. The RT prescription [1] was derived from the gravitational path integral by Lewkowycz and Maldacena [2]; Dong [3] extended the argument to Rényi entropies, showing that the refined Rényi entropy is the area of a backreacting cosmic brane of tension $T_q = (q - 1)/(4qG_N)$. Explicit brane solutions for spherical regions are hyperbolic black holes [4]. Freedman and Headrick reformulated RT as a max-flow problem (“bit threads”) [5]; explicit geodesic flows were constructed in [6]. Holography at a finite radial cutoff is related to $T\bar{T}$ -type deformations [7, 8]. On the mathematical side, the theory of complex dimensions of fractal strings and sets [15, 16] explains log-periodic oscillations in spectral and geometric asymptotics through poles of zeta functions, the multifractal formalism [17] describes measures with a continuum of local scaling exponents, and DSI and its log-periodic signatures are reviewed in [18]. In two-dimensional CFT, the entanglement spectrum is known in closed form [9, 10].

Entanglement across fractal boundaries has been studied before, and our work builds directly on these results. Hama, Lida and Severini showed that in the toric code the entanglement across a fractal boundary of Hausdorff dimension D obeys $S/p = \gamma \leq 1/D$ [26]. Faraji Asstaneh computed the entanglement entropy of regions on fractal spaces with heat-kernel methods for a free scalar [24]. He found a leading divergence controlled by the spectral dimension and log-periodic corrections originating in the complex dimensions of the fractal, including for a Cantor-string entangling boundary analysed with Lapidus’ zeta function, and he proposed a hyperscaling-violating holographic description. Akal found log-periodic corrections for boundaries with the structure of finitely ramified self-similar graphs [25]. Log-periodic entanglement also arises in aperiodic spin chains [28, 29], and generalised area laws governed by the boundary dimension have been observed on fractal lattices [26, 27]. Holographic entanglement for non-smooth entangling curves, including singular points formed by logarithmic spirals, was studied in [31, 32, 33, 30], and discrete scale invariance has been studied in holography from several angles [34, 35, 36].

Results. In contrast to [24], where the space itself is fractal and the field theory is free, we keep the bulk and the boundary smooth (Poincaré AdS dual to a CFT on $\mathbb{R}^{d-1,1}$) and let only the entangling surface be fractal, working throughout with the RT and cosmic-brane prescriptions. Within this setting our results are:

1. *Holographic box counting* (Theorem 1). The identity $-\varepsilon\partial_\varepsilon S_A = (R/4G_N)\text{Vol}(\gamma_A \cap \{z = \varepsilon\})$ holds exactly. In the linearised problem the bulk acts on the boundary profile by an explicit positive smoothing kernel, and for a Weierstrass-type boundary we prove $-\varepsilon\partial_\varepsilon S_A \asymp (R^{d-1}/4G_N)N_\varepsilon(\partial A)$ with N_ε the box-counting function.
2. *DSI locking* (Theorem 2). We adapt the renewal-theoretic proof of the lattice/non-lattice dichotomy for Minkowski contents [15, 23, 44, 45] to the RT cross-section content. The new inputs are the bulk extension of boundary similarities and a near-boundary decoupling property (Lemma 4). The complex dimensions governing S_A are then the zeros of the Moran function $1 - \sum_j r_j^s$, and the bulk dynamics, linear or not, enters only through the residues.
3. *Non-local counterterm* (Proposition 4.1). Building on the area renormalisation of entangling surfaces [12, 38], we construct the counterterm that renders S_A finite. We show (Proposition 4.2) that local counterterms fail because cancellation requires inverting the radial Euler operator $-z\partial_z$ on a lattice of complex exponents. This is related in spirit to the scale-derivative refinement of Liu and Mezei [37].
4. *Marginal complex dimensions* (Theorem 3). Within a distance-zeta approximation, we give a necessary and sufficient condition for bounded log-periodic terms in the renormalised entropy, together with an explicit lattice example.

5. *Cosmic branes and multifractality* (Proposition 6.1). Dong already noted that the refined Rényi entropy is a thermodynamic entropy with respect to the modular Hamiltonian [3, 41, 42]. Combining this with the multifractal formalism gives $\tilde{S}_q = f(\alpha_q) \ln(1/\varepsilon)$. The identity itself is elementary; its content is that in Einstein gravity the shape of f is fixed by the bulk, a consistency check that in $d = 2$ reproduces the known Calabrese–Lefevre spectrum [9, 10].
6. *A spectral fingerprint* (Section 8) of marginal complex dimensions, namely half-integer log-harmonics, in lattice fractal-string resonators. It follows from the explicit formulas of fractal-string spectral theory [15] and tests the zeta-function structure shared with our holographic analysis, not holography itself.

We are explicit about assumptions. Lemma 4 is proven in the linearised problem and supported, but not proven, in the nonlinear one. Self-affine boundaries are not covered by Theorem 2. The scale-transport equation of Section 7 is an organising conjecture.

Organisation. Section 2 fixes conventions. Section 3 derives the fractal area law. Section 4 treats self-similar boundaries, complex dimensions and counterterms. Section 5 analyses marginal complex dimensions. Section 6 turns to Rényi entropies and cosmic branes. Section 7 discusses bit threads and scale transport, and Section 8 the experimental fingerprint. We conclude in Section 9. Numerical methods are collected in Appendix A.

2 Geometry: Poincaré AdS and the radial resolution

We work on the $t = 0$ slice of Poincaré AdS_{d+1} ,

$$ds^2 = \frac{R^2}{z^2} (dz^2 + dx^2 + d\vec{y}^2), \quad \vec{y} \in \mathbb{R}^{d-2}, \quad (2)$$

a copy of hyperbolic space $\mathbb{H}^d$. Boundary similarities $y \mapsto rOy + t$ extend to the bulk isometries $(y, z) \mapsto (rOy + t, rz)$.

3 Fractal area law

3.1 Setup

Let $A \subset \mathbb{R}^{d-1}$ be a region whose boundary ∂A has upper and lower box (Minkowski) dimension $D_\partial \in [d-2, d-1]$, defined through the box-counting function $N_\delta(\partial A)$, the minimal number of δ -cubes covering ∂A :

$$D_\partial = \lim_{\delta \rightarrow 0} \frac{\ln N_\delta(\partial A)}{\ln(1/\delta)}. \quad (3)$$

We use box rather than Hausdorff dimension throughout: it is the box-counting function that enters the entropy, and the two dimensions coincide for the self-similar sets of Section 4.

For explicit computations we use *graph-type* boundaries,

$$\partial A = \{x = f_0(y_1)\} \times [0, L_\perp]^{d-3}, \quad f_0(y_1) = A_0 \sum_{n \geq 0} a^n \cos(k_0 b^n y_1), \quad b > 1, \quad b^{-1} < a < 1, \quad (4)$$

periodic in y_1 with period $Y = 2\pi/k_0$ (we take b integer). The graph of the Weierstrass function f_0 has box dimension $D = 2 + \ln a / \ln b \in (1, 2)$ [20], so $D_\partial = D + d - 3$. For $d = 3$ the entangling surface is a fractal curve in the plane. Fractal curves therefore live naturally in $\text{CFT}_3/\text{AdS}_4$, while in AdS_5 the construction (4) produces a fractal “cylinder”.

3.2 An exact radial identity

Near the boundary we parametrise γ_A as a graph $x = f(\vec{y}, z)$ over $(\vec{y}, z)$ for $z < z_0$. The induced metric is $h_{ab} = (R^2/z^2)(\delta_{ab} + \partial_a f \partial_b f)$ in coordinates $(\vec{y}, z)$, so $\sqrt{h} = (R/z)^{d-1} \sqrt{1 + |\partial f|^2}$ with $|\partial f|^2 = |\nabla_y f|^2 + f_z^2$. Define the *cross-section content*

$$\mathcal{A}(z) = \int d^{d-2}y \sqrt{1 + |\nabla_y f|^2 + f_z^2}. \quad (5)$$

Lemma 1 (Radial identity). *With a radial cutoff at $z = \varepsilon < z_0$,*

$$\begin{aligned} \text{Area}_\varepsilon(\gamma_A) &= \text{Area}(\gamma_A \cap \{z > z_0\}) + R^{d-1} \int_\varepsilon^{z_0} dz z^{1-d} \mathcal{A}(z), \\ -\varepsilon \partial_\varepsilon S_A &= \frac{R}{4G_N} \text{Vol}(\gamma_A \cap \{z = \varepsilon\}), \end{aligned} \quad (6)$$

where $\text{Vol}(\gamma_A \cap \{z = \varepsilon\}) = (R/\varepsilon)^{d-2} \mathcal{A}(\varepsilon)$ is the proper $(d-2)$ -volume of the cross-section.

Proof. The first equation is the area integral in the coordinates above. The second follows by differentiation: $-\varepsilon \partial_\varepsilon \text{Area}_\varepsilon = R^{d-1} \varepsilon^{2-d} \mathcal{A}(\varepsilon) = R (R/\varepsilon)^{d-2} \mathcal{A}(\varepsilon)$. $\square$

Lemma 1 is exact and independent of any fractal structure; it is an immediate consequence of RT in Poincaré coordinates, and the interpretation of the scale dependence of entanglement entropy as an RG flow goes back to [37]. The physics lies entirely in the small- z behaviour of $\mathcal{A}(z)$, which we now determine.

3.3 The linearised problem and the radial kernel

Varying the area functional gives the RT equation

$$\partial_z \left(\frac{z^{1-d} f_z}{\sqrt{1 + |\partial f|^2}} \right) + \nabla_y \cdot \left(\frac{z^{1-d} \nabla_y f}{\sqrt{1 + |\partial f|^2}} \right) = 0, \quad (7)$$

whose linearisation about the plane $f = 0$, standard in studies of the shape dependence of holographic entanglement [39, 40], is

$$f_{zz} - \frac{d-1}{z} f_z + \nabla_y^2 f = 0, \quad f(\vec{y}, 0) = f_0(\vec{y}). \quad (8)$$

For a Fourier mode $f = \varphi(z) e^{ik \cdot y}$, (8) becomes the modified Bessel equation in the variable $x = |k|z$ after the substitution $\varphi = x^\nu u(x)$, $\nu = d/2$. Its solution decaying as $z \rightarrow \infty$ and normalised by $\varphi(0) = 1$ is

$$\phi_d(x) = \frac{x^\nu K_\nu(x)}{2^{\nu-1} \Gamma(\nu)}, \quad \nu = \frac{d}{2}. \quad (9)$$

For $d = 3$ it is elementary, $\phi_3(x) = (1+x)e^{-x}$. The two regimes of (9) follow from the standard expansions of K_ν :

$$\phi_d(x) = 1 - \frac{x^2}{2(d-2)} + \cdots + c_d x^d (1 + O(x^2)) \quad [\times \ln x \text{ for even } d], \quad x \rightarrow 0, \quad (10)$$

$$\phi_d(x) = \frac{\sqrt{\pi/2}}{2^{\nu-1} \Gamma(\nu)} x^{\frac{d-1}{2}} e^{-x} (1 + O(x^{-1})), \quad x \rightarrow \infty. \quad (11)$$

The small- x expansion follows from $K_\nu = \frac{\pi}{2} \frac{I_{-\nu} - I_\nu}{\sin \nu \pi}$ and the reflection formula $\Gamma(\nu) \Gamma(1-\nu) = \pi / \sin \nu \pi$. The large- x form is Hankel's asymptotic expansion. Thus features of wavenumber k are transmitted essentially unchanged to depth $z \ll 1/k$ and are exponentially suppressed for $z \gg 1/k$. The first non-local datum enters at order x^d , the normalisable mode.

Proposition 3.1 (Radial smoothing kernel). *The solution of (8) is $f(\cdot, z) = K_z * f_0$ with*

$$K_z(\vec{y}) = \frac{\Gamma(d-1)}{\pi^{\frac{d-2}{2}} \Gamma(\frac{d}{2})} \frac{z^d}{(|\vec{y}|^2 + z^2)^{d-1}}, \quad \int_{\mathbb{R}^{d-2}} K_z = 1, \quad K_z > 0. \quad (12)$$

Proof. In $\mathbb{R}^n$ the Fourier transform of $(1 + |y|^2)^{-a}$ is proportional to $|k|^{a-n/2} K_{a-n/2}(|k|)$. With $n = d - 2$ and $a = d - 1$ we get $a - n/2 = d/2 = \nu$, which matches (9) after rescaling $y \rightarrow y/z$. Dimensional analysis fixes the power z^d , and the normalisation follows from $\phi_d(0) = 1$. For $d = 3$ one checks directly that $(1 - z\partial_z) \frac{z}{\pi(y^2 + z^2)} = \frac{2z^3}{\pi(y^2 + z^2)^2}$, the image of $e^{-|k|z}(1 + |k|z)$. $\square$

The kernel (12) makes the slogan “radial depth = resolution” precise. The linearised RT surface at depth z is the boundary profile averaged with a positive, unit-mass kernel of width z , and its tail $\int_{|y|>\ell} K_z = O((z/\ell)^d)$ quantifies how weakly distant boundary data influence the surface. This is the basis of Lemma 4 below.

3.4 Content asymptotics for Weierstrass boundaries

Write $x = k_0 z$ and $k_n = k_0 b^n$. For (4), orthogonality of the Fourier modes on one period gives

$$\|\partial_{y_1} f(\cdot, z)\|_{L^2}^2 = \frac{Y}{2} A_0^2 k_0^2 \sum_{n \geq 0} (ab)^{2n} \phi_d(b^n x)^2, \quad \|f_z(\cdot, z)\|_{L^2}^2 = \frac{Y}{2} A_0^2 k_0^2 \sum_{n \geq 0} (ab)^{2n} \phi'_d(b^n x)^2. \quad (13)$$

Since $a = b^{D-2}$, the growth factor is $ab = b^{D-1} > 1$.

Lemma 2 (Log-periodic sums). *Let $h \geq 0$ be continuous on $(0, \infty)$, bounded near 0, decaying exponentially at ∞ , and not identically zero. Then*

$$\Psi_h(x) := \sum_{n \in \mathbb{Z}} (ab)^{2n} h(b^n x) = x^{-2(D-1)} P_h(\log_b x), \quad (14)$$

with P_h continuous, 1-periodic and strictly positive. The one-sided sum over $n \geq 0$ differs from Ψ_h by $O(1)$ as $x \rightarrow 0$.

Proof. The sum converges because $(ab)^{2n} \rightarrow 0$ for $n \rightarrow -\infty$, where h is bounded, and because h decays exponentially for $n \rightarrow +\infty$. Shifting the summation index gives $\Psi_h(bx) = (ab)^{-2} \Psi_h(x) = b^{-2(D-1)} \Psi_h(x)$, so $x^{2(D-1)} \Psi_h(x)$ is invariant under $x \mapsto bx$, i.e. periodic in $\log_b x$. It is continuous, and positive because some term is positive. The terms with $n < 0$ sum to at most $\sup h \sum_{n < 0} (ab)^{2n} < \infty$. $\square$

Applying Lemma 2 with $h = \phi_d^2$ and $h = \phi'_d{}^2$ gives $\|\partial_{y_1} f\|_2, \|f_z\|_2 \asymp z^{1-D}$ as $z \rightarrow 0$. To control $\mathcal{A}$, which involves L^1 norms, we use that both $\partial_{y_1} f(\cdot, z)$ and $f_z(\cdot, z)$ are *Hadamard-lacunary* trigonometric series in y_1 (frequency ratio $b > 1$). For such series all L^p norms, $p < \infty$, are equivalent with constants depending only on b [21]: $B_b \|g\|_2 \leq \|g\|_1 \leq \sqrt{Y} \|g\|_2$.

Proposition 3.2 (Content bounds). *For the linearised RT surface of (4) there are constants $0 < c_1 \leq c_2$ and $z_* > 0$ such that*

$$c_1 z^{(d-2)-D\partial} \leq \mathcal{A}(z) \leq c_2 z^{(d-2)-D\partial}, \quad 0 < z \leq z_*. \quad (15)$$

Proof. From $\max(1, |\partial_{y_1} f|) \leq \sqrt{1 + |\partial f|^2} \leq 1 + |\partial_{y_1} f| + |f_z|$ and integration over a period, $L_\perp^{d-3} \max(Y, \|\partial_{y_1} f\|_1) \leq \mathcal{A} \leq L_\perp^{d-3} (Y + \|\partial_{y_1} f\|_1 + \|f_z\|_1)$. Lacunarity and Lemma 2 turn both bounds into $\asymp z^{1-D}$ once $z^{1-D} \gg 1$. Finally, $z^{1-D} L_\perp^{d-3} \asymp z^{(d-2)-D\partial}$ in units where the transverse box count is $(L_\perp/z)^{d-3}$. $\square$

3.5 Holographic box counting

Theorem 1 (Holographic box counting). *Let γ_A be the RT surface of A and $\mathcal{A}$ its cross-section content (5).*

$$(i) \text{ (Exact.) } -\varepsilon \partial_\varepsilon S_A = \frac{R}{4G_N} \text{Vol}(\gamma_A \cap \{z = \varepsilon\}) = \frac{R^{d-1}}{4G_N} \varepsilon^{2-d} \mathcal{A}(\varepsilon).$$

(ii) (Scaling.) If (15) holds with $D_\partial > 0$, then

$$\frac{R^{d-1} c_1}{4G_N D_\partial} \varepsilon^{-D_\partial} (1 + o(1)) \leq S_A \leq \frac{R^{d-1} c_2}{4G_N D_\partial} \varepsilon^{-D_\partial} (1 + o(1)). \quad (16)$$

(iii) (Log-periodic refinement.) If $z^{D_\partial - (d-2)} \mathcal{A}(z) = P(\ln z) + o(1)$ with P continuous and periodic of period $\ln b$, $P(t) = \sum_m P_m e^{2\pi i m t / \ln b}$, then

$$S_A = \frac{R^{d-1}}{4G_N} \varepsilon^{-D_\partial} \tilde{P}(\ln \varepsilon) + o(\varepsilon^{-D_\partial}), \quad (17)$$

$$\tilde{P}(t) = \int_0^\infty e^{-D_\partial u} P(t + u) du, \quad \tilde{P}_m = \frac{P_m}{D_\partial - 2\pi i m / \ln b}.$$

In particular, the mean coefficient is reduced by $1/D_\partial$ relative to the smooth-boundary normalisation. (A factor $1/D$ also appears in the toric-code bound of [26], for different reasons.)

(iv) (Box counting.) For the linearised Weierstrass model, $\mathcal{A}(\varepsilon) \asymp \varepsilon^{d-2} N_\varepsilon(\partial A)$, hence

$$-\varepsilon \partial_\varepsilon S_A \asymp \frac{R^{d-1}}{4G_N} N_\varepsilon(\partial A). \quad (18)$$

Proof. (i) is Lemma 1. For (ii), insert (15) into (6): the integral $R^{d-1} \int_\varepsilon^{z_*} z^{1-d} z^{(d-2)-D_\partial} dz = R^{d-1} (\varepsilon^{-D_\partial} - z_*^{-D_\partial}) / D_\partial$, and the remaining contributions are ε -independent. For (iii), substitute $z = \varepsilon e^u$: $\int_\varepsilon z^{-1-D_\partial} P(\ln z) dz = \varepsilon^{-D_\partial} \int_0^\infty e^{-D_\partial u} P(\ln \varepsilon + u) du$, and the $o(1)$ remainder contributes $o(\varepsilon^{-D_\partial})$. Mode by mode, $\int_0^\infty e^{-D_\partial u} e^{i\omega(t+u)} du = e^{i\omega t} / (D_\partial - i\omega)$. For (iv), the box-counting function of the graph of f_0 satisfies $N_\delta \asymp \delta^{-D}$ [20], so $N_\varepsilon(\partial A) \asymp \varepsilon^{-D} (L_\perp / \varepsilon)^{d-3}$. Comparing with Proposition 3.2 and (i) gives (18). $\square$

Remark. In the linearised model, (13) and Lemma 2 show that the L^2 content is *exactly* log-periodic up to $O(1)$, and the hypothesis of (iii) holds for $\|\nabla f\|_2^2$. For the L^1 content $\mathcal{A}$ itself, the hypothesis of (iii) follows from the renewal structure established in Section 4. Linearisation is not uniformly controlled at small z , since slopes grow like z^{1-D} . For self-similar boundaries, however, Section 4 shows that the scaling of $\mathcal{A}$ is fixed by symmetry independently of linearisation.

3.6 Numerical test

We computed $\mathcal{A}(z)$ for the linearised surface with $b = 3$, six modes, $A_0 = k_0 = 1$ and $d = 3$, on 2^{15} points per period, and fitted power laws over the window $3/k_{\max} \leq z \leq 0.3/k_0$ (about 2.4 decades). The results are in Table 1 and Fig. 1.

The deviations, 0.03–0.08, are systematic rather than statistical, and four effects account for them. (a) *The smooth pole.* The content contains a constant term c_s (the “1” under the square root, i.e. the exponent $\omega = d - 2$ of a smooth boundary). A pure power-law fit absorbs it and underestimates $|1 - D|$, as seen in the $\mathcal{A}$ column. The bias is largest when the two exponents are close; for $D = 1.1$ the model is moreover pre-asymptotic, since $ab = 3^{0.1}$ makes the slopes $O(1)$ throughout the window. (b) *Infrared subtraction.* The regulated entropy is $C\varepsilon^{-D} - C'$ with $C' > 0$ from the upper limit of the window. Its logarithmic slope, $-D/(1 - (C'/C)\varepsilon^D)$,

D	$\mathcal{A}$ exponent (num.)	$1 - D$	S_A exponent (num.)	$-D$
1.1	-0.114	-0.100	-1.154	-1.100
1.3	-0.248	-0.300	-1.260	-1.300
1.5	-0.439	-0.500	-1.422	-1.500
1.7	-0.654	-0.700	-1.621	-1.700
1.9	-0.871	-0.900	-1.836	-1.900

Table 1: Fitted scaling exponents of the cross-section content and of the regulated entropy (linearised Weierstrass model, $d = 3$, $b = 3$, six modes) against the predictions of Theorem 1.

is steeper than $-D$, which explains the steeper entropy exponents at small D . (c) *Log-periodic modulation*. Its relative amplitude is about 1% (Section 4). Over a window of about four periods this biases a straight-line fit at the 10^{-3} level, a subdominant effect. (d) *Truncation and non-homogeneity*. With finitely many modes the content saturates below $z \sim 1/k_{\max}$, and the integrand $\sqrt{1 + |\partial f|^2}$ restores exact affine scaling only for $|\partial f| \gg 1$. Including the smooth pole and three log-periodic harmonics in the fit (Section 4.5) reduces the discrepancy at $D = 1.5$ from 0.061 to 0.024, with relative residual 1.5×10^{-3} .

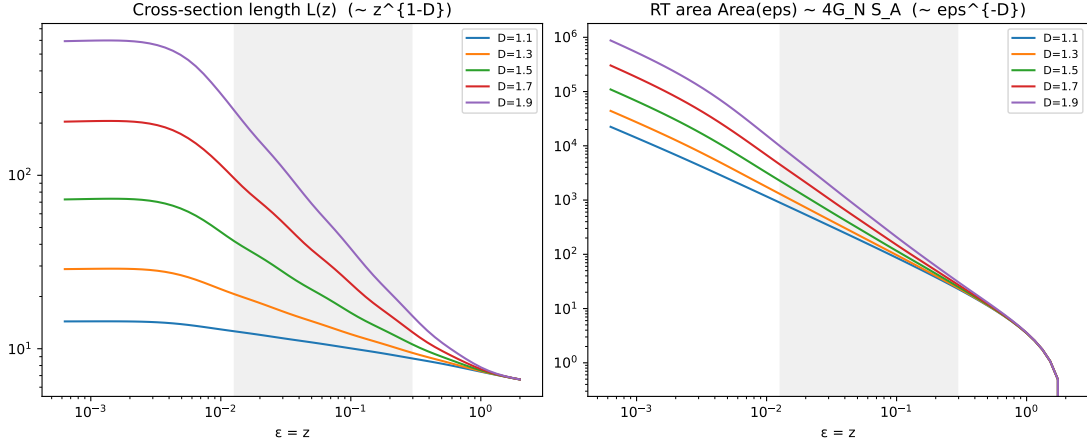


Figure 1: Cross-section content $\mathcal{A}(z)$ (left) and regulated RT area (right) for the linearised Weierstrass model. The shaded band marks the fit window.

4 Discrete scale invariance and complex dimensions

4.1 Self-similar boundaries and the bulk similarity group

Definition 4.1. Let $\{S_j\}_{j=1}^N$ be similarities of $\mathbb{R}^{d-1}$, $S_j(y) = r_j O_j y + t_j$ with $0 < r_j < 1$ and $O_j \in O(d-1)$, satisfying the open set condition. Their attractor $K = \bigcup_j S_j(K)$ has box and Hausdorff dimension equal to the similarity dimension D_∂ , the unique real solution of $\sum_j r_j^{D_\partial} = 1$ [19, 20]. Set $\lambda_j = \ln(1/r_j)$ and let $\Lambda \subset (\mathbb{R}, +)$ be the subgroup generated by $\{\lambda_j\}$. The system is *lattice* if $\Lambda = (\ln b)\mathbb{Z}$ for some $b > 1$, and *non-lattice* otherwise, in which case Λ is dense.

Lemma 3 (Bulk extension). *Each S_j extends to the isometry $\hat{S}_j(y, z) = (S_j(y), r_j z)$ of (2). If γ is the RT surface of a region B , then $\hat{S}_j(\gamma)$ is the RT surface of $S_j(B)$, and the cross-section contents satisfy $\mathcal{A}_{\hat{S}_j(\gamma)}(z) = r_j^{d-2} \mathcal{A}_\gamma(z/r_j)$.*

Proof. The metric $R^2(dz^2 + dy^2)/z^2$ is invariant under $(y, z) \mapsto (rOy + t, rz)$. Isometries preserve areas and homology, hence minimality. The cross-section of $\hat{S}_j(\gamma)$ at depth z is the image of the

cross-section of γ at depth z/r_j under S_j , whose coordinate $(d-2)$ -volume scales by r_j^{d-2} . $\square$

4.2 Near-boundary decoupling

Lemma 4 (Near-boundary decoupling). *There exist $\eta > 0$ and $C < \infty$ with the following property. If two regions A, A' coincide in a ball $B_\ell(p)$, then for $z \leq \ell$ the cross-section contents of γ_A and $\gamma_{A'}$ over $B_{\ell/2}(p)$ differ by at most $C(z/\ell)^\eta$ times either one.*

We do not have a proof of Lemma 4 for the nonlinear problem with fractal data. Three independent lines of evidence support it.

1. *Linearised problem (proven).* By Proposition 3.1, a change of boundary data outside $B_\ell(p)$ alters f at (y, z) , $y \in B_{\ell/2}(p)$, by at most $\|\delta f_0\|_\infty \int_{|y'| > \ell/2} K_z$, which is $O((z/\ell)^d)$. Hence $\eta = d$.
2. *Fefferman–Graham structure (smooth case).* For smooth ∂A the embedding of an extremal surface has the near-boundary expansion $X^i(\sigma, z) = X_0^i(\sigma) + z^2 X_2^i(\sigma) + \dots + z^d X_d^i(\sigma) + \dots$, with logarithms for even d . Here $X_2^i \propto K^i$ (the trace of the extrinsic curvature of ∂A) and all coefficients below X_d are local functionals of ∂A [12]. X_d is the first coefficient not fixed by local data, so non-local information enters at relative order z^d , consistent with $\eta = d$.
3. *Comparison principle.* By entanglement wedge nesting [14], $\gamma_{A \cap A'}$ and $\gamma_{A \cup A'}$ bound γ_A and $\gamma_{A'}$ from either side. Nesting confines the deviation qualitatively; a quantitative estimate would require barrier surfaces controlled near the boundary.

For fractal ∂A the Fefferman–Graham expansion does not apply (curvatures are unbounded), and in the nonlinear problem η need not equal d . Theorem 2 only uses $\eta > 0$.

4.3 DSI locking

The mathematical core of the following theorem is not new. That the complex dimensions of a self-similar set are the zeros of the Moran function, and that its Minkowski content is log-periodic in the lattice case and converges in the non-lattice case, is due to Lapidus and van Frankenhuysen, Falconer, Gatzouras and Lalley [15, 44, 45, 23, 46], and log-periodicity as a signature of DSI is reviewed in [18]. We adapt their renewal-theoretic argument to the RT cross-section content. The new inputs are Lemma 3, which transports the boundary self-similarity to the bulk, and Lemma 4, which controls the non-locality of the minimal-surface problem.

For a region A with $\partial A = K$ and z_0 small, define the content zeta function

$$\zeta_{\mathcal{A}}(s) = \int_0^{z_0} z^{s-(d-1)} \mathcal{A}(z) dz, \quad (19)$$

normalised so that a term $C z^{(d-2)-\omega}$ in $\mathcal{A}$ produces a pole $C/(s-\omega)$. Set

$$M(s) = 1 - \sum_{j=1}^N r_j^s. \quad (20)$$

Theorem 2 (DSI locking). *Let $\partial A = K$ be a self-similar set as above, assume Lemma 4, and let $\mathcal{A}$ be locally bounded. Then:*

- (i) $\zeta_{\mathcal{A}}$ converges for $\operatorname{Re} s > D_\partial$ and continues meromorphically to $\operatorname{Re} s > D_\partial - \eta'$ for some $\eta' > 0$. There

$$\zeta_{\mathcal{A}}(s) = \frac{\zeta_g(s) + E(s)}{M(s)}, \quad (21)$$

with E entire and ζ_g holomorphic. Its poles there lie in the zero set $\mathfrak{D} = \{s : M(s) = 0\}$.

(ii) On the line $\operatorname{Re} s = D_\partial$,

$$\mathfrak{D} \cap \{\operatorname{Re} s = D_\partial\} = D_\partial + i\Lambda^\perp, \quad \Lambda^\perp = \{t \in \mathbb{R} : e^{it\lambda} = 1 \ \forall \lambda \in \Lambda\}, \quad (22)$$

so $\Lambda^\perp = (2\pi/\ln b)\mathbb{Z}$ in the lattice case and $\Lambda^\perp = \{0\}$ otherwise. All these zeros are simple.

(iii) The location of the poles depends only on the scaling ratios $\{r_j\}$. The bulk dynamics, linear or nonlinear, enters only through the residues $C_m = (\zeta_g(s_m) + E(s_m))/M'(s_m)$ at $s_m = D_\partial + it_m$, $t_m \in \Lambda^\perp$.

(iv) In the lattice case, $z^{D_\partial-(d-2)}\mathcal{A}(z) = P(\ln z) + o(1)$ with P periodic of period $\ln b$, whose Fourier frequencies lie in $\Lambda^\perp$. In the non-lattice case the limit is a constant.

Proof. Step 1 (Renewal equation). Decompose the near-boundary part of γ_A into pieces lying over neighbourhoods of $S_j(K)$. By Lemma 3, the RT surface of $S_j(A)$ is $\hat{S}_j(\gamma_A)$. Near $S_j(K)$ but away from the finitely many junction points where different copies meet, A and $S_j(A)$ coincide in balls of radius comparable to the distance ρ to the junctions. Lemma 4 then bounds the difference of local contents by $(z/\rho)^\eta$. Summing over dyadic shells $\rho = 2^k z$, the shell at radius ρ carries content $\lesssim z^{d-2}(\rho/z)^{D_\partial}$, so the total error is

$$g(z) \lesssim z^{d-2} \sum_{k \geq 0} 2^{k(D_\partial - \eta)} \Big|_{2^k z \lesssim 1} = O(z^{(d-2)-D_\partial + \min(\eta, D_\partial)}) \quad (23)$$

(up to a logarithm when $\eta = D_\partial$). Hence

$$\mathcal{A}(z) = \sum_{j=1}^N r_j^{d-2} \mathcal{A}(z/r_j) + g(z), \quad g(z) = O(z^{(d-2)-D_\partial + \eta'}), \quad \eta' > 0. \quad (24)$$

Step 2 (Mellin transform). For $\operatorname{Re} s > D_\partial$ the bound (15) makes (19) converge. Substituting $u = z/r_j$,

$$\int_0^{z_0} z^{s-(d-1)} r_j^{d-2} \mathcal{A}(z/r_j) dz = r_j^s \int_0^{z_0/r_j} u^{s-(d-1)} \mathcal{A}(u) du = r_j^s \zeta_{\mathcal{A}}(s) + E_j(s), \quad (25)$$

where E_j is an integral over the compact range $[z_0, z_0/r_j]$ and hence entire. Transforming (24) gives (21) with $E = \sum_j E_j$ and ζ_g the transform of g , which is holomorphic for $\operatorname{Re} s > D_\partial - \eta'$ by the bound on g . This proves (i).

Step 3 (Zeros on the critical line: duality). This step is the standard argument of [15], phrased in the language of Pontryagin duality. Write $s = D_\partial + it$ and $w_j = r_j^{D_\partial} > 0$, so $\sum_j w_j = 1$. Then $M(s) = 0$ reads

$$\sum_j w_j e^{-it\lambda_j} = 1. \quad (26)$$

The left side is a convex combination of points of the unit circle, and 1 is an extreme point of the closed unit disk. Equality therefore holds if and only if $e^{-it\lambda_j} = 1$ for every j . Each $\chi_t(\lambda) = e^{it\lambda}$ is a character of $(\mathbb{R}, +)$, and every character of $\mathbb{R}$ has this form ($\hat{\mathbb{R}} \cong \mathbb{R}$). A character is trivial on Λ if and only if it is trivial on its generators. The solutions are therefore exactly $t \in \Lambda^\perp$, the annihilator of Λ in $\hat{\mathbb{R}}$. If $\Lambda = (\ln b)\mathbb{Z}$ then $\Lambda^\perp = (2\pi/\ln b)\mathbb{Z}$. If Λ is dense, a continuous character trivial on Λ is trivial on $\mathbb{R}$, so $\Lambda^\perp = \{0\}$. These zeros are simple because $M'(s) = \sum_j r_j^s \lambda_j$, which on $D_\partial + i\Lambda^\perp$ equals $\sum_j w_j \lambda_j > 0$.

Step 4 (Rigidity). By (i) and Step 3, the poles on the critical line are determined by Λ alone. The bulk dynamics enters (24) only through g and hence only through $\zeta_g + E$ in the numerator of (21), which proves (iii). Equivalently, $\Gamma = \Lambda$ acts on $\ln z$ by translations; a Γ -invariant function is a function on $\mathbb{R}/\Lambda \cong \mathbb{T}$, whose Fourier dual is the discrete group $\Lambda^\perp$. No

deformation that preserves the similarity data can move the frequencies continuously, which is the precise sense in which they are locked.

Step 5 (Asymptotics). Set $\ell = -\ln z$ and $F(\ell) = e^{-(D_\partial - (d-2))\ell} \mathcal{A}(e^{-\ell})$. Then (24) becomes the renewal equation $F(\ell) = \sum_j w_j F(\ell - \lambda_j) + \tilde{g}(\ell)$, whose kernel is a probability measure with $\tilde{g}$ directly Riemann integrable by the bound on g . The lattice and non-lattice renewal theorems [22, 23] give, respectively, convergence of F to a $(\ln b)$ -periodic function or to a constant. Its Fourier coefficients are the residues C_m of (iii). $\square$

Remark. (1) The real part of the critical line is locked as well: D_∂ is the unique real zero of M , which is strictly decreasing on $\mathbb{R}$. Nonlinear corrections do *not* renormalise the dimension of a self-similar boundary. (2) Self-affine boundaries such as (4) are not covered, because anisotropic scalings are not isometries of (2). In the linearised problem, however, $f \mapsto af$ is a symmetry of (8), so a renewal structure holds for the L^2 content (Lemma 2). The effective exponent $D_{\text{fit}} \neq D$ found numerically for the Weierstrass model reflects the non-homogeneity of $\sqrt{1 + |\partial f|^2}$ and mode truncation (Section 3.6), not a renormalisation of D . (3) Off the critical line, $\mathfrak{D}$ may contain further zeros with $D_\partial - \eta' < \text{Re } s < D_\partial$. They produce subleading log-periodic terms, and their imaginary parts need not lie in $\Lambda^\perp$.

4.4 The non-local counterterm

On the cutoff surface $\Sigma_c = \{z = z_c\}$ the cross-section $\partial\gamma_c = \gamma_A \cap \Sigma_c$ has proper volume $V(z_c) = (R/z_c)^{d-2} \mathcal{A}(z_c)$. Let $\mathfrak{D}_+$ denote the set of poles of $\zeta_{\mathcal{A}}$ with $\text{Re } \omega > 0$, including the smooth pole $\omega = d - 2$ when present, and assume the asymptotic expansion

$$\mathcal{A}(z) = \sum_{\omega \in \mathfrak{D}_+} C_\omega z^{(d-2)-\omega} + \mathcal{A}_0(z), \quad (27)$$

where $\mathcal{A}_0$ contributes at most marginally (Section 5). Lemma 1 then gives

$$\text{Area}_{z_c} = R^{d-1} \sum_{\omega \in \mathfrak{D}_+} \frac{C_\omega}{\omega} z_c^{-\omega} + (\text{marginal}) + (\text{finite}). \quad (28)$$

Let Π_+ be the projection of the expansion of V onto the exponents in $\mathfrak{D}_+$, so $\Pi_+ V(z) = R^{d-2} \sum_{\omega \in \mathfrak{D}_+} C_\omega z^{-\omega}$.

Proposition 4.1 (Non-local counterterm). *The functional*

$$I_{ct}(z_c) = -\frac{R}{4G_N} \int_{z_c}^\infty \frac{dz}{z} \Pi_+ V(z) = -\frac{R}{4G_N} (-z_c \partial_{z_c})^{-1} \Pi_+ V(z_c) \quad (29)$$

cancels all divergences with $\text{Re } \omega > 0$ in $S_A(z_c)$. Equivalently, I_{ct} is the decaying solution of the first-order flow equation

$$z_c \partial_{z_c} I_{ct} = \frac{R}{4G_N} \Pi_+ V(z_c). \quad (30)$$

For a smooth boundary ($\mathfrak{D}_+ = \{d - 2\}$) it reduces to the standard leading counterterm $-\frac{R}{4G_N(d-2)} \text{Vol}(\partial\gamma_c)$.

Proof. For $\text{Re } \omega > 0$, $\int_{z_c}^\infty z^{-\omega-1} dz = z_c^{-\omega}/\omega$, so (29) equals $-\frac{R^{d-1}}{4G_N} \sum C_\omega z_c^{-\omega}/\omega$. This cancels (28) divided by $4G_N$. The operator $(-z\partial_z)^{-1}$ acts diagonally as $z^{-\omega} \mapsto z^{-\omega}/\omega$, and (30) follows by differentiation. In the smooth case $\Pi_+ V = V \propto z^{-(d-2)}$. $\square$

Remark. In $d = 3$, with $\mathcal{A} = L$, (29) can be written as $-\frac{R^2}{4G_N} z_c^{-1} (1 - z_c \partial_{z_c})^{-1} \Pi_+ L(z_c)$, because $(1 - z\partial_z)z^{1-\omega} = \omega z^{1-\omega}$. The operator $(1 - z\partial_z)^{-1}$ acts on the coordinate content L , *not* on the proper volume V . On V the correct operator is $(-z\partial_z)^{-1}$.

Proposition 4.2 (Origin of the failure of local counterterms). *Let ∂A be a lattice self-similar set with $z^{D_\partial-(d-2)}\mathcal{A}(z) \rightarrow P(\ln z)$ and P non-constant. Then there is no constant κ such that $S_A(z_c) - \kappa \frac{R}{4G_N} V(z_c)$ remains bounded as $z_c \rightarrow 0$. If P is constant, the unique admissible value is $\kappa = 1/D_\partial$, which depends on A .*

Proof. By Theorem 1(iii), the leading divergence of S_A is $\frac{R^{d-1}}{4G_N} z_c^{-D_\partial} \tilde{P}(\ln z_c)$, while $\kappa \frac{R}{4G_N} V = \kappa \frac{R^{d-1}}{4G_N} z_c^{-D_\partial} P(\ln z_c)(1 + o(1))$. Boundedness of the difference requires $\tilde{P} = \kappa P$. By (17) this means $P_m/s_m = \kappa P_m$ for all m , with $s_m = D_\partial - 2\pi i m / \ln b$. Since the s_m are distinct, at most one P_m can be non-zero. Because P is real, $P_{-m} = \overline{P_m}$, so that mode must be $m = 0$, and then $\kappa = 1/D_\partial$. $\square$

The mathematical root is now transparent. A divergence of the form $\int_{z_c} \frac{dz}{z} V(z)$ is cancelled by inverting the radial Euler operator $-z\partial_z$ on V . When V is a single power with a universal exponent, as for smooth boundaries, this inversion is multiplication by the universal constant $1/(d-2)$, which is local. For a fractal it is the Fourier multiplier $m \mapsto 1/s_m$ on the lattice of complex dimensions. That multiplier depends on the non-universal D_∂ and reweights each log-periodic mode by a different complex factor. A functional evaluated on a single cutoff slice cannot implement it. Adding local curvature invariants of $\partial\gamma_c$ only adds further periodic functions of $\ln z_c$ with universal coefficients, and does not remove the A -dependence of the zero mode. We state this last point for families with continuously varying D_∂ without a general proof.

Remark (Relation to existing renormalisation schemes). Covariant counterterms localised on $\partial\gamma_c$ were constructed for smooth entangling surfaces in [38]; Proposition 4.2 explains why that construction cannot be carried over unchanged to fractal surfaces. Liu and Mezei removed area-law divergences by differential operators in the region size, e.g. $(R\partial_R - 1)$ in $d = 3$ [37]. For a self-similar region the analogous operator $(R\partial_R - D_\partial)$ annihilates only the $m = 0$ mode. Removing all modes would require the formal product $\prod_m (R\partial_R - s_m)$, the differential counterpart of the inverse operator in (29).

Remark (Scheme dependence). A cutoff reparametrisation $\varepsilon \mapsto \varepsilon h(\ln \varepsilon)$ with h periodic can absorb any log-periodic modulation of the leading divergence; this point was made in [24]. Our statements refer to the geometric RT cutoff $z = \varepsilon$, with only constant rescalings $\varepsilon \mapsto \lambda\varepsilon$ regarded as scheme changes. This is the same convention that makes logarithmic coefficients universal. Under constant rescalings the moduli $|C_m|$ are invariant and only their phases change.

Remark (Finite cutoff). At finite z_c the Dirichlet wall is physical and no counterterm is added. The full $z_c^{-D_\partial} \tilde{P}(\ln z_c)$ dependence is then an observable of the cutoff theory. The flow form (30) is local in the flow parameter. Under the cutoff-AdS/ $T\bar{T}$ dictionary, with deformation parameter $\mu \propto G_N z_c^d$ [7, 8], one has $z_c \partial_{z_c} = d\mu \partial_\mu$. The counterterm therefore obeys a local first-order equation along the $T\bar{T}$ flow even though it is not a local functional at fixed μ . For $d > 2$ this identification is heuristic.

4.5 Numerical illustration

For the linearised Weierstrass model with $D = 1.5$, $b = 3$ and seven modes, fitting $\mathcal{A}(z) = c_s + z^{1-D} [c_0 + \sum_{m=1}^3 (\alpha_m \cos + \beta_m \sin)(2\pi m \ln z / \ln 3)]$ gives $D_{\text{fit}} = 1.524$ with relative residual 1.5×10^{-3} . The leading complex dimension is at $\text{Im } s = \pm 5.72 = \pm 2\pi / \ln 3$ with $|C_1/c_0| = 8.6 \times 10^{-3}$, and the higher harmonics are suppressed further. Folding $(\mathcal{A} - c_s)z^{D-1}$ by $\log_3 z$ collapses the data onto a single periodic curve. With the counterterm (29) built from the fitted residues, the renormalised entropy drifts by 5 (in units of $R^2/4G_N$) across the window, while the bare area changes by 1.2×10^4 . The local counterterm $-RV/4G_N$ leaves a residual $\approx (1/D - 1)c_0 z_c^{-D}$ of order -6×10^3 (Fig. 2).

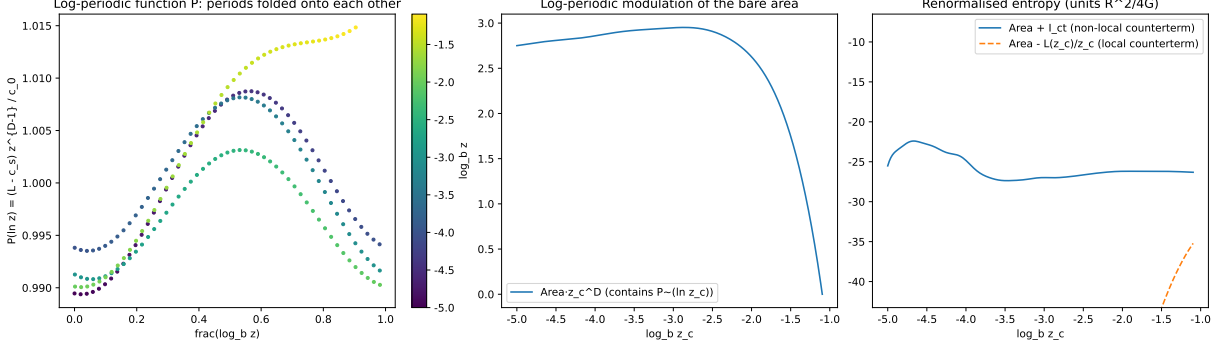


Figure 2: Left: log-periodic function P folded over periods. Middle: bare area times z_c^D . Right: renormalised entropy with the non-local counterterm (29) (solid) and with the local counterterm (dashed).

5 Marginal complex dimensions

Poles with $\text{Re } \omega > 0$ produce power divergences, which Proposition 4.1 removes. Poles on the line $\text{Re } \omega = 0$ are *marginal*. A real pole at $\omega = 0$ gives a logarithm, as for corners [31, 32, 33], and non-real poles $\omega = it$ give terms $\varepsilon^{-it}/(it)$ that neither diverge nor decay. Such terms survive any minimal subtraction. In this section we determine when they occur.

5.1 Thread landing and the distance zeta function

To analyse marginal terms we need more than the content $\mathcal{A}(z)$: we need to know where on the boundary the entanglement measured at depth z resides. Bit threads (Section 7) provide this. For a half-space $A = \{x < 0\}$, the geodesic flow of Proposition 7.1 consists of semicircles of radius u crossing $\gamma_A = \{x = 0\}$ at depth $z = u$ and landing at distance u from ∂A . Equating the flux through $\gamma_A \cap \{u < z < u + du\}$ with the number of threads landing in the corresponding boundary strip shows that the landing density on A is exactly

$$\varrho(y) = \frac{R^{d-1}}{4G_N} \text{dist}(y, \partial A)^{-(d-1)}. \quad (31)$$

For a general region we adopt (31) as the leading near-boundary approximation, justified when ∂A is approximately planar on the scale $\text{dist}(y, \partial A)$. This is the situation at depth z by Proposition 3.1, since there ∂A is resolved only down to scale z . Then

$$S_A(\varepsilon) \simeq \frac{R^{d-1}}{4G_N} \int_{A, \text{dist} > \varepsilon} \text{dist}(y, \partial A)^{-(d-1)} d^{d-1}y = \frac{R^{d-1}}{4G_N} \int_{\varepsilon}^{\delta} u^{-(d-1)} dV_A(u) + O(1), \quad (32)$$

where $V_A(u) = |\{y \in A : \text{dist}(y, \partial A) < u\}|$ is the inner tube volume. For smooth ∂A , $V_A(u) = \text{Area}(\partial A) u + O(u^2)$, which reproduces the area-law coefficient exactly.

Equation (32) identifies the holographic entropy with the *relative distance zeta function* of Lapidus, Radunović and Žubrinić [16],

$$\zeta_{\partial A, A}(s) = \int_{A, \text{dist} < \delta} \text{dist}(y, \partial A)^{s-(d-1)} d^{d-1}y, \quad (33)$$

evaluated at $s = 0$ and regulated at $\text{dist} = \varepsilon$. Its poles are the complex dimensions of the relative fractal drum $(\partial A, A)$. A pole ω with residue c_ω contributes $c_\omega \varepsilon^{-\omega}/\omega$ to (32) for $\omega \neq 0$, and $c_0 \ln(1/\varepsilon)$ for $\omega = 0$. This is the precise sense in which the regulated value of the distance zeta function at $s = 0$ is the AdS boundary contact term: the RT integrand $u^{-(d-1)} dV_A$ is exactly the distance-zeta integrand at $s = 0$. The tube formulas of Lapidus and collaborators [15, 16, 47]

expand V_A as a sum over the same poles, so the divergence structure of S_A is read off from the tube formula of ∂A . Equation (32) is an approximation. Corrections of the type $g(z)$ in Theorem 2 modify residues but, by that theorem, not pole locations on the critical line.

5.2 The criterion

For self-similar sprays and tilings the distance zeta function factorises [16, 47]:

$$\zeta_{\partial A, A}(s) = \frac{\zeta_{\text{gen}}(s)}{M(s)} + (\text{entire}), \quad M(s) = 1 - \sum_j m_j r_j^s. \quad (34)$$

Here ζ_{gen} is the distance zeta function of the generator and m_j counts copies with ratio r_j . In the lattice case, $r_j = b^{-k_j}$ with $\gcd(k_j) = 1$, set $x = b^{-s}$ and define the *Moran polynomial*

$$\mathcal{P}(x) = \sum_j m_j x^{k_j} - 1, \quad M(s) = -\mathcal{P}(b^{-s}). \quad (35)$$

Theorem 3 (Marginal criterion). *Assume (32) and (34). The holographic entropy contains a bounded, non-decaying log-periodic term, i.e. $\zeta_{\partial A, A}$ has a non-real pole on $\text{Re } s = 0$, if and only if at least one of the following holds:*

- (i) $\mathcal{P}$ has a root $x_* = e^{-i\theta}$ on the unit circle, $\theta \notin 2\pi\mathbb{Z}$, and $\zeta_{\text{gen}}(i\theta/\ln b) \neq 0$ (counting the order of a possible zero of ζ_{gen} against the multiplicity of the root);
- (ii) ζ_{gen} has a non-real pole on $i\mathbb{R}$ that is not cancelled by a zero of $1/M$.

Proof. Poles of the quotient (34) on $i\mathbb{R}$ are either zeros of M not cancelled by zeros of the numerator, or poles of ζ_{gen} not cancelled by poles of M . Since M is entire, it has no poles, so the second case is exactly (ii). For the first, $M(it) = 0$ iff $\mathcal{P}(e^{-it \ln b}) = 0$, i.e. iff $\mathcal{P}$ has a root on the unit circle. The root $x = 1$ is excluded because $\mathcal{P}(1) = \sum_j m_j - 1 \geq 1$ for a genuine fractal ($\sum m_j \geq 2$). By (32) a simple pole $\omega = it$ with residue c_ω contributes $c_\omega \varepsilon^{-it}/(it)$, which is bounded and does not decay. A pole at $s = 0$ itself contributes $\ln(1/\varepsilon)$ instead. $\square$

Corollary 5.1 (Single-ratio fractals). *If all ratios are equal, $r_j = 1/b$ with N copies, then $\mathcal{P}(x) = Nx - 1$ has its only root at $1/N$, inside the unit disk, so (i) never holds. In particular, for the Koch curve ($N = 4$, $b = 3$), whose generator is polygonal and has only real poles, the marginal line carries only the real pole $s = 0$. The entropy then contains a universal logarithm,*

$$S_A \supset \lambda \frac{R^{d-1}}{4G_N} \ln \frac{1}{\varepsilon}, \quad \lambda = \frac{\text{Res}_{s=0} \zeta_{\text{gen}}}{M(0)} = \frac{\text{Res}_{s=0} \zeta_{\text{gen}}}{1 - N}, \quad (36)$$

but no bounded oscillation. The residue at $s = 0$ of a polygonal generator comes from its corners, so this is a renormalised corner logarithm [31]: the corners of all generations are resummed by the factor $1/(1 - N)$.

5.3 An explicit lattice example

Consider the lattice string with one copy of ratio 2^{-1} and one of ratio 2^{-5} . Its Moran polynomial factorises as

$$\mathcal{P}(x) = x^5 + x - 1 = (x^2 - x + 1)(x^3 + x^2 - 1). \quad (37)$$

To verify, expand the right-hand side: $x^5 + x^4 - x^2 - x^4 - x^3 + x + x^3 + x^2 - 1 = x^5 + x - 1$.

The first factor is the cyclotomic polynomial Φ_6 , with roots $x = e^{\pm i\pi/3}$ on the unit circle. The cubic has one real root $x_0 \simeq 0.7549$, giving $D = \ln(1/x_0)/\ln 2 \simeq 0.4057$. Its other two roots have modulus $x_0^{-1/2} \simeq 1.151$, giving $\text{Re } s \simeq -0.2028$. The complex dimensions are therefore

$$s \in \left\{ D + \frac{2\pi i m}{\ln 2} \right\} \cup \left\{ \frac{2\pi i}{\ln 2} \left(m \pm \frac{1}{6} \right) \right\} \cup \left\{ -0.2028 + \dots \right\}, \quad m \in \mathbb{Z}. \quad (38)$$

The marginal family produces a bounded oscillation whose fundamental period in $\ln \varepsilon$ is $6 \ln 2$.

The mechanism is combinatorial. Let c_K be the number of copies at log-depth K , i.e. with ratio 2^{-K} . It obeys $c_K = c_{K-1} + c_{K-5}$, whose generating function is $-1/\mathcal{P}(x)$. The factor Φ_6 gives c_K an exactly periodic, undamped component of period 6 in K , on top of the growing component x_0^{-K} . The growing component, weighted by r^D , produces the power divergence. The periodic component enters the marginal (scale-invariant, r^0 -weighted) term, where it cannot be removed by any counterterm targeting $\text{Re } \omega > 0$. More generally, a cyclotomic factor Φ_n of $\mathcal{P}$ gives a period $n \ln b$. Unit-circle roots that are not roots of unity would give quasi-periodic marginal terms; we have not classified when they occur.

Numerical check. In an exactly solvable renewal model with generator $g(u) = ue^{-u}$, so that $\zeta_g = \Gamma(s)$ has only real poles, we computed $S(\varepsilon) = \int_{\varepsilon}^1 z^{-2} \mathcal{A} dz$ exactly and subtracted all poles with $\text{Re } \omega > 0$ and the logarithm $\ln(1/\varepsilon)/M(0)$. For the Koch-type system the remainder is constant to 9×10^{-4} . For the $(2^{-1}, 2^{-5})$ string it oscillates with peak-to-peak amplitude 0.16 and period $6 \ln 2$, agreeing with the Mellin prediction $\sum_{\text{Re } \omega=0} c_{\omega} \varepsilon^{-\omega}/\omega$ to 3×10^{-4} (Fig. 3).

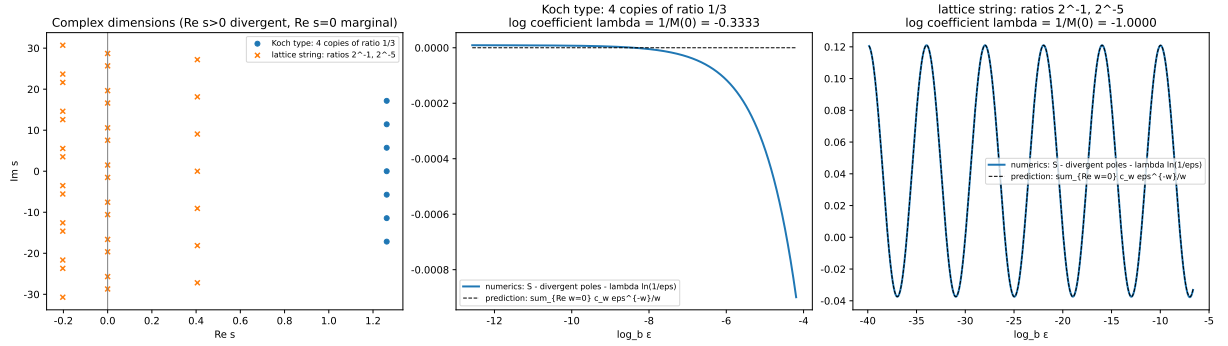


Figure 3: Left: complex dimensions of the Koch-type system and the $(2^{-1}, 2^{-5})$ lattice string. Middle and right: renormalised entropy after subtracting divergent poles and the logarithm (solid) against the prediction from marginal complex dimensions (dashed).

6 Rényi entropies, cosmic branes and multifractality

6.1 Dong's area law and its thermodynamic reading

For a holographic CFT, Dong showed that [3]

$$\tilde{S}_q \equiv q^2 \partial_q \left(\frac{q-1}{q} S_q \right) = \frac{\text{Area}(\mathcal{C}_q)}{4G_N}, \quad (39)$$

where $\mathcal{C}_q$ is a cosmic brane with tension $T_q = (q-1)/(4qG_N)$. The brane backreacts on the bulk by creating a conical deficit $\Delta\phi = 2\pi(q-1)/q$, and the replica derivation follows Lewkowycz and Maldacena [2]. Dong also noted that $\tilde{S}_q = -\partial F_q / \partial T$ with $F_q = -\frac{1}{q} \ln \text{Tr} \rho^q$ and $T = 1/q$. In other words, $\tilde{S}_q$ is the thermodynamic entropy of the modular Hamiltonian $K = -\ln \rho$ at temperature $1/q$, i.e. the von Neumann entropy of the escort state $\rho_q = \rho^q / \text{Tr} \rho^q$. He cited the thermodynamic formalism of Beck and Schlögl [41] and Baez [42]; see also [43].

6.2 Cosmic-brane areas as a singularity spectrum

The thermodynamic formalism of multifractals [17, 41] applies verbatim once $\text{Tr} \rho_A^q$ scales with the cutoff. We record the resulting identity.

Proposition 6.1 (Brane area and singularity spectrum). *Let $\mathcal{L} \rightarrow \infty$ be a large parameter, for instance $\mathcal{L} = \ln(1/\varepsilon)$, and suppose*

$$\ln \text{Tr} \rho_A^q = -\tau(q) \mathcal{L} + o(\mathcal{L}) \quad (40)$$

pointwise on an open interval I of q on which τ is differentiable. Then for every $q \in I$,

$$\frac{\tilde{S}_q}{\mathcal{L}} \rightarrow f(\alpha_q) \equiv q\alpha_q - \tau(q), \quad \frac{\langle K \rangle_{\rho_q}}{\mathcal{L}} \rightarrow \alpha_q \equiv \tau'(q), \quad (41)$$

so that, by (39), $\text{Area}(\mathcal{C}_q)/4G_N = f(\alpha_q) \mathcal{L} (1 + o(1))$.

Proof. $\Phi(q) := \ln \text{Tr} \rho^q$ is convex in q , since it is a log-sum-exp of functions linear in q . Pointwise convergence of the differentiable convex functions $\Phi/\mathcal{L} \rightarrow -\tau$ on I , with differentiable limit, implies convergence of their derivatives on I [48, Thm. 25.7], so $-\Phi'/\mathcal{L} \rightarrow \tau'$. Since $\frac{q-1}{q} S_q = -\frac{1}{q} \Phi$, we get $\tilde{S}_q = q^2 \partial_q (-\Phi/q) = \Phi - q\Phi'$, hence $\tilde{S}_q/\mathcal{L} \rightarrow -\tau + q\tau'$. Finally $\langle K \rangle_{\rho_q} = -\Phi'(q)$. $\square$

The proof is two lines, and we claim no novelty for the identity: it is the standard Legendre structure of the thermodynamic formalism transcribed through (39). Its use here is interpretive. The tension of the brane selects a Hölder exponent α_q , and the brane area counts the eigenvalues of ρ_A near $e^{-\alpha_q \mathcal{L}}$. The multifractal spectrum of the entanglement spectrum is therefore a *geometric* quantity in the bulk, namely the area of a one-parameter family of cosmic branes.

6.3 Einstein gravity: the shape of f is fixed by the bulk

For a spherical (or planar) entangling surface the backreacted geometry is a hyperbolic black hole [4, 3]:

$$ds^2 = f(r) d\tau^2 + \frac{dr^2}{f(r)} + r^2 dH_{d-1}^2, \quad f(r) = r^2 - 1 - \frac{r_h^d - r_h^{d-2}}{r^{d-2}} \quad (R=1). \quad (42)$$

Its temperature is $T(r_h) = f'(r_h)/4\pi = (dr_h^2 - (d-2))/(4\pi r_h)$, with $T(1) = T_0 = 1/2\pi$. Smoothness of the q -fold cover, equivalently the deficit $2\pi(q-1)/q$ in the quotient, requires $T(x_q) = T_0/q$:

$$dx_q^2 - \frac{2}{q} x_q - (d-2) = 0 \quad \implies \quad x_q = \frac{1 + \sqrt{1 + d(d-2)q^2}}{dq}. \quad (43)$$

The brane sits at the horizon, so $\tilde{S}_q = x_q^{d-1} S_1$, and [4]

$$\frac{S_q}{S_1} = \frac{q}{2(q-1)} \left(2 - x_q^{d-2} (1 + x_q^2) \right). \quad (44)$$

With $\mathcal{L} = S_1$, Proposition 6.1 gives $f(\alpha_q) = x_q^{d-1}$ and $\alpha_q = \partial_q [(q-1)S_q/S_1]$. We verified $q^2 \partial_q (\frac{q-1}{q} S_q)/S_1 = x_q^{d-1}$ numerically to 10^{-8} for $d = 2, 3, 4$.

For a fractal entangling surface the near-boundary part of the brane at depth z sees a locally planar surface (Proposition 3.1). To leading order the Rényi entropies then factorise:

$$S_q^{\text{div}} \simeq \left(\frac{S_q}{S_1} \right)_{\text{grav}} \times S_1^{\text{div}}, \quad S_1^{\text{div}} \propto N_\varepsilon(\partial A). \quad (45)$$

The geometry of ∂A fixes only the overall scale $\mathcal{L}$; the shape of $f(\alpha)$ is fixed by the gravitational theory. This factorisation is an approximation that we have not proven for fractal surfaces.

The case $d = 2$. Here $x_q = 1/q$. Writing $\mathcal{L} = \ln(\ell/\varepsilon)$ for an interval,

$$\tau(q) = \frac{c}{6} \left(q - \frac{1}{q} \right), \quad \alpha_q = \frac{c}{6} \left(1 + \frac{1}{q^2} \right), \quad f(\alpha_q) = \frac{c}{3q}, \quad f(\alpha) = \sqrt{\frac{c}{3} \left(2\alpha - \frac{c}{3} \right)}, \quad \alpha \geq \frac{c}{6}. \quad (46)$$

This is the exponential growth rate of the Calabrese–Lefevre eigenvalue density [9]. It can be recovered from the BCFT description of Cardy and Tonni [10], as follows. The reduced density matrix is the annulus of width $W = 2\mathcal{L}$, whose open-channel eigenvalues are

$$\lambda_h = \frac{e^{-(2\pi^2/W)(h-c/24)}}{Z(W)}, \quad (47)$$

with h the dimensions of boundary-condition-changing operators. The closed channel gives $\ln Z = cW/12 + O(1)$. With the Cardy density $\rho(E) \sim e^{2\pi\sqrt{cE/6}}$, $E = h - c/24$, the eigenvalues at $\alpha = -\ln \lambda/\mathcal{L}$ have $E = \mathcal{L}^2(\alpha - c/6)/\pi^2$, and their number is $e^{2\pi\sqrt{cE/6}} = e^{\mathcal{L}\sqrt{(2c/3)(\alpha-c/6)}}$, which is (46). We present this as a consistency check between (39), [10] and [9], not as a new result. Numerically, the saddle-point prediction is approached with deviation 8×10^{-2} ($W = 400$) and 2.4×10^{-2} ($W = 2000$), in units of $c/6$ (Fig. 4).

6.4 Sparse spectra and the $1/c$ tension

Equation (47) gives Schmidt weights of the form $p \propto e^{-\alpha\Delta}$ with $\alpha = 2\pi^2/W$. Suppose, as a model assumption, that for a self-similar region the modular Hamiltonian factorises across generations, each generation j being an annulus of log-width $\lambda_j = \ln(1/r_j)$. The cascade weights are then

$$p_j \propto n_j e^{-(2\pi^2/\lambda_j)\Delta_j}, \quad \sum_j p_j^q r_j^{-\tau(q)} = 1. \quad (48)$$

For a dense (Cardy) spectrum the shape of f is again universal. Non-trivial multifractality requires a *sparse* spectrum of Δ_j . For Ising boundary-condition-changing operators, $\Delta \in \{0, 1/16, 1/2\}$, with $b = 3$ we find $p = (0.754, 0.245, 9 \times 10^{-5})$ and $D_1 = 0.51$.

This exposes a tension. Classical (large- c) holography implies Cardy-like density and hence a universal shape of f fixed by x_q . Sparse spectra, needed for non-trivial cascade weights, are not described by Einstein gravity. In a holographic theory, cascade-type multifractality can therefore appear only at $O(c^0)$: in bulk quantum corrections [49], or in superpositions of fixed-area states [50, 51]. We have not derived p_j from first principles beyond the factorised model.

Finally, for a lattice cascade the partition function obeys a renewal equation whose poles are at $\tau(q) + 2\pi im/\ln b$. The argument of Theorem 2, Step 3, with weights $w_j(q) = p_j^q r_j^{-\tau(q)} > 0$ summing to 1, shows that the imaginary parts are q -independent. We confirmed this numerically for $r = (1/2, 1/4)$, where the oscillation has period $\ln 2$ for $q = -2, 0, 2, 4$. For a non-lattice cascade the oscillation is not periodic (Fig. 5).

7 Bit threads and scale transport

7.1 Flux depth distribution and geodesic flows

Freedman and Headrick showed that [5]

$$S_A = \max_v \int_A v, \quad \nabla_\mu v^\mu = 0, \quad |v| \leq \frac{1}{4G_N}, \quad (49)$$

and that every maximal flow saturates the bound normal to γ_A . Explicit flows were constructed in [6]. We adapt their geodesic construction to the planar case.

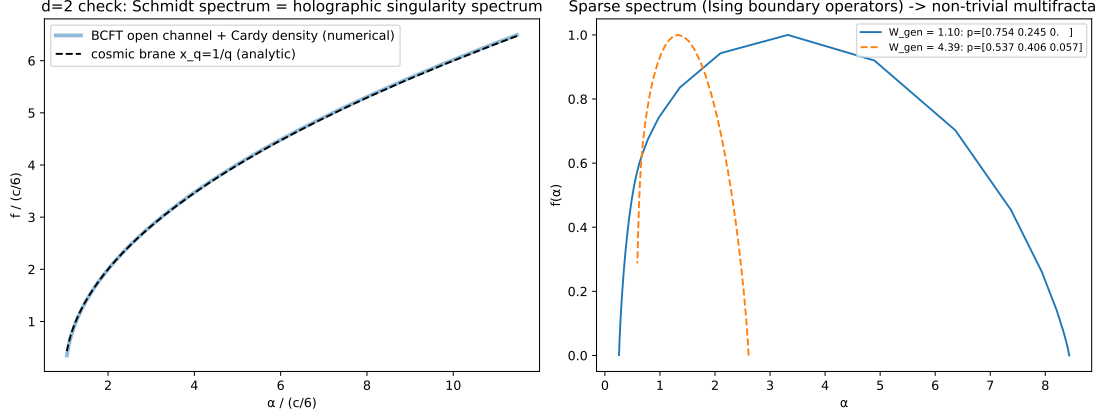


Figure 4: Left: singularity spectrum from BCFT open-channel weights with Cardy density (numerical) against the cosmic-brane result (46). Right: cascade spectra from Ising boundary operators.

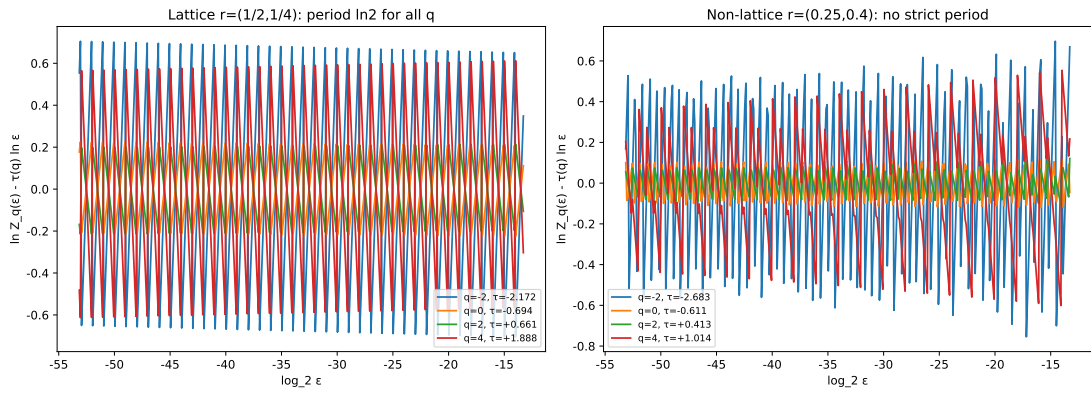


Figure 5: $\ln Z_q(\varepsilon) - \tau(q) \ln \varepsilon$ for lattice (left) and non-lattice (right) binomial cascades.

Proposition 7.1 (Thread depth distribution and geodesic flow). *(i) The flux through $\gamma_A \cap \{z_1 < z < z_2\}$ is the same for every maximal flow and equals $\int_{z_1}^{z_2} n(z) dz$, with*

$$n(z) = \frac{R^{d-1}}{4G_N} z^{1-d} \mathcal{A}(z) = \frac{R}{4G_N} \frac{V(z)}{z}, \quad (50)$$

independent of the flow.

(ii) The flux lost when the cutoff is raised, $-\partial_{z_c} \Phi(z_c) = n(z_c)$, is related to the counterterm (29) by $\partial_{z_c} I_{ct} = \Pi_+ n(z_c)$.

(iii) For a half-space $A = \{x < 0\}$, the vector field tangent to the semicircles $x^2 + z^2 = u^2$ (at fixed $\vec{y}$), with

$$|v| = \frac{\sin^{d-1} \theta}{4G_N}, \quad \sin \theta = \frac{z}{u}, \quad (51)$$

is divergence-free and satisfies $|v| \leq 1/4G_N$, with equality exactly on $\gamma_A = \{x = 0\}$.

Proof. (i) On γ_A a maximal flow has $v \cdot n = 1/4G_N$ [5], so the flux through a portion of γ_A is its area divided by $4G_N$. The area element is given by Lemma 1. (ii) The first statement follows from (i). The second is (30) rewritten using $n = (R/4G_N)V/z$.

(iii) On the slice (2), $g_{\mu\nu} = (R^2/z^2)\delta_{\mu\nu}$ and $\sqrt{g} = (R/z)^d$, so

$$\nabla_\mu v^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} v^\mu). \quad (52)$$

Semicircles orthogonal to the boundary are geodesics of $\mathbb{H}^d$. In the (x, z) plane the coordinate unit tangent, oriented from A to $\bar{A}$, is $t = (z, -x)/u$. The corresponding proper unit vector is $\hat{v} = (z/R)t$, because $|t|_g = (R/z)|t|_\delta$. With $v = \frac{1}{4G_N}(z/u)^{d-1}\hat{v}$, the only non-zero components are

$$\sqrt{g} v^x = \frac{R^{d-1}}{4G_N} \frac{z}{u^d}, \quad \sqrt{g} v^z = -\frac{R^{d-1}}{4G_N} \frac{x}{u^d}, \quad u = \sqrt{x^2 + z^2}. \quad (53)$$

Hence

$$\partial_x \left(\frac{z}{u^d} \right) - \partial_z \left(\frac{x}{u^d} \right) = -\frac{dxz}{u^{d+2}} + \frac{dxz}{u^{d+2}} = 0, \quad (54)$$

and $\nabla_\mu v^\mu = 0$; the $\vec{y}$ -components vanish identically. The norm is $|v| = \frac{1}{4G_N}(z/u)^{d-1} \leq \frac{1}{4G_N}$, with equality iff $z = u$, i.e. $x = 0$. There $\hat{v} = \partial_x$ in proper normalisation, so $v \cdot n = 1/4G_N$ and the bound is saturated on γ_A . (The computation was also checked symbolically.) $\square$

Part (iii) is the planar limit of the construction of [6]. Parts (i)–(ii) give the counterterm a direct interpretation: I_{ct} removes exactly the threads whose depth distribution lies in the divergent complex-dimension modes, and $S_{\text{ren}} = \lim_{z_c \rightarrow 0} \int_{z_c} (1 - \Pi_+) n dz$.

Rényi threads (proposal). Since $\tilde{S}_q$ is the area of $\mathcal{C}_q$ in the backreacted geometry g_q , max-flow/min-cut on the time slice of g_q gives $\tilde{S}_q = \max \int_A v_q$, *provided* $\mathcal{C}_q$ is minimal, not merely extremal, on that slice. Under this assumption the loss rate of q -threads per e -fold of z is $f(\alpha_q) \partial_\ell \mathcal{L}$ with $\ell = \ln(1/z)$, by Proposition 6.1. We have not established the minimality assumption in general.

7.2 A scale-transport equation

The preceding results share one structure: a quantity at log-scale $\ell = \ln(\ell_0/z)$ is a weighted sum of copies of itself at shifted scales, plus a source. We propose this structure as an organising principle.

Conjecture 1 (Scale transport). *For a self-similar ∂A with similarities S_j , $\lambda_j = \ln(1/r_j)$, there exist state-dependent weights $p_j > 0$, $\sum_j p_j = 1$, and a source σ_ℓ of lower growth, such that the measure ρ_ℓ describing the entanglement (thread landing) density resolved at scale ℓ satisfies*

$$\rho_\ell = \sum_j p_j (S_j)_\# \rho_{\ell-\lambda_j} + \sigma_\ell. \quad (55)$$

Here $(S_j)_\#$ is the push-forward, so the right side is the Hutchinson operator [19] acting along a renewal in ℓ .

Taking q -th moments over a partition at resolution $e^{-\ell}$ gives $Z_q(\ell) = \sum_j p_j^q Z_q(\ell - \lambda_j) + G_q(\ell)$. Its Laplace transform has poles at the complex mass exponents $\tau(q) + i\Lambda^\perp$. If Conjecture 1 holds, the results of this paper are its specialisations:

Specialisation	Result
geometric weights $p_j = r_j^{D_0}$ ($q = 0$ data)	complex dimensions, Theorem 2
poles with $\text{Re} > 0$	non-local counterterm, Proposition 4.1
poles on $\text{Re} = 0$	logarithms and bounded oscillations, Theorem 3
Legendre transform in q	cosmic-brane areas, Proposition 6.1
$-\partial_\ell$ of the first moment	thread loss rate, Proposition 7.1

The conjecture is established only in two cases. The first is the geometric case, where it reduces to the renewal equation (24), subject to Lemma 4. The second is the factorised BCFT model of the previous section, where it holds by assumption. We do not derive non-geometric weights p_j from first principles, and (55) should be read as a framework rather than a theorem.

8 Experimental fingerprint: lattice fractal-string resonators

The renewal and zeta-function structure behind marginal complex dimensions is not specific to holography. It governs the spectral asymptotics of fractal strings [15], produces log-periodic terms in the thermodynamics of fields on fractals [52], and has been discussed as an observable signature of discrete scale invariance [53]. This suggests a laboratory test of the structure that our holographic analysis shares with fractal-string theory. We stress that such a test probes the zeta-function mechanism of Section 5, not holography itself.

Geometry. Consider a lattice fractal string generated by one copy of ratio $r_1 = 1/3$ and two copies of ratio $r_2 = 1/9$, realised as a collection of one-dimensional resonators (for instance transmission-line or acoustic resonators) of lengths $\ell_w = \ell_0 r_w$, where r_w is the product of the ratios along the word w . Its geometric zeta function is $\zeta_L(s) = \sum_w r_w^s = 1/M(s)$ with

$$M(s) = 1 - 3^{-s} - 2 \cdot 9^{-s} = -\mathcal{P}(3^{-s}), \quad \mathcal{P}(x) = 2x^2 + x - 1 = (2x - 1)(x + 1). \quad (56)$$

The root $x = 1/2$ gives $D = \ln 2 / \ln 3 \simeq 0.6309$. The root $x = -1$, a zero of the cyclotomic factor $\Phi_2(x) = x + 1$, places a marginal family exactly on the imaginary axis,

$$s_m^{(0)} = \frac{(2m+1)\pi i}{\ln 3}, \quad m \in \mathbb{Z}, \quad (57)$$

and $\mathcal{P}$ has no other roots.

Observable. With Dirichlet ends, the resonance frequencies are $f_{w,k} = kv/(2\ell_w)$, $k \geq 1$, where v is the phase velocity. We use the heat-kernel-smoothed mode count

$$\Theta(u) = \sum_w \sum_{k \geq 1} \exp\left(-\frac{k^2}{(ur_w)^2}\right), \quad u = \frac{2\ell_0 f}{v}, \quad (58)$$

which can be computed directly from a measured list of resonances. Its Mellin transform is

$$\int_0^\infty u^{-s-1} \Theta(u) du = \frac{1}{2} \Gamma(s/2) \zeta(s) \zeta_L(s), \quad (59)$$

where $\zeta(s)$ is the Riemann zeta function, so that as $u \rightarrow \infty$

$$\begin{aligned} \Theta(u) &= \frac{\sqrt{\pi}}{2} \zeta_L(1) u + \sum_{m \in \mathbb{Z}} A_m^{(D)} u^{D+2\pi i m / \ln 3} + \frac{1}{4} + \sum_{m \in \mathbb{Z}} A_m^{(0)} u^{(2m+1)\pi i / \ln 3} + \dots, \\ A &= \frac{\Gamma(s/2) \zeta(s)}{2M'(s)} \Big|_{\text{pole}}. \end{aligned} \quad (60)$$

The constant is $\frac{1}{2} \cdot 2 \cdot \zeta(0)/M(0) = 1/4$. The poles of $\Gamma(s/2)$ at $s = -2, -4, \dots$ are cancelled by the trivial zeros of ζ , so the remainder decays faster than any power of u .

Fingerprint. After the Weyl term is removed, a Fourier analysis in $\log_3 u$ separates two families of lines. The D family gives *integer* harmonics (an integer number of cycles per $\log_3$ unit), with envelope growing as $u^D \propto f^D$. The marginal family gives *half-integer* harmonics with bounded, non-decaying amplitude. Numerically, the residual $\Theta - \text{Weyl} - D$ family $-1/4$ agrees with the marginal sum in (60) to 2×10^{-6} . Its peak-to-peak amplitude is 0.059 modes, the fundamental has $|A_0^{(0)}| = 0.0145$, and the pattern repeats every factor 9 in frequency (Fig. 6).

Requirements and caveats. Two full periods of the marginal oscillation require a frequency ratio of 81. For example, with $\ell_0 = 10$ cm and $v = 1.2 \times 10^8$ m/s the band is 0.60–48.6 GHz. Only branches with $r_w \geq 3^{-4}$ have modes in this band, and there are $1 + 1 + 3 + 5 + 11 = 21$ of them. The mean Weyl spacing of resonances is $v/(2\ell_0 \zeta_L(1)) = \frac{4}{9} v/(2\ell_0) \simeq 0.27$ GHz, independent of frequency, so near the top of the band a quality factor of at least a few hundred is needed merely to separate neighbouring resonances. Exact degeneracies between equal-length branches must be counted with their multiplicity. Because the marginal amplitude is only 0.059 modes, the count must be exact: a single missed or spurious resonance in the analysed band is comparable to the signal. Fabrication tolerances on the lengths shift resonances and must be included in an error budget, which we have not carried out. Finally, for *uncoupled* branches the spectrum is the union of harmonic ladders, and (60) follows from the lengths alone. The experiment then tests the explicit formula and the realisability of the measurement, not new physics. A physically non-trivial test would use a coupled network or a single cavity with a lattice-self-similar boundary, for which the smoothed counting function is not additive. In that case the pole *locations* of (60) are still fixed by $M(s)$, but the residues are not.

9 Discussion and limitations

We summarise what is established, what is assumed, and what remains open.

1. *Near-boundary decoupling.* Lemma 4 is proven in the linearised RT problem, with $\eta = d$, and is consistent with the Fefferman–Graham structure and with entanglement wedge nesting [14]. Nesting, however, controls the deviation only qualitatively. A non-linear proof would require quantitative barrier estimates for the minimal-surface equation in smooth AdS with non-rectifiable (fractal) boundary data. Theorems 2 and 3 inherit this assumption.

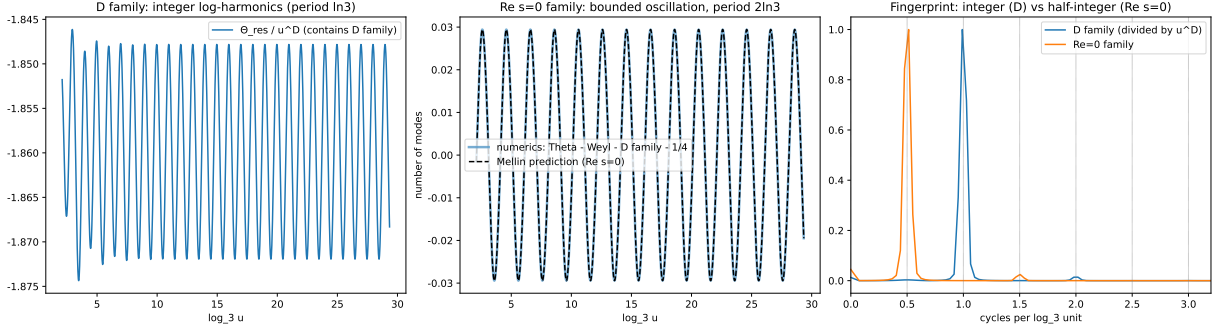


Figure 6: Lattice fractal-string resonator. Left: residual heat trace divided by u^D , showing the integer log-harmonics of the D family. Middle: residual after removing the D family, compared with the marginal prediction from (60). Right: Fourier spectra in $\log_3 u$, with integer lines for the D family and half-integer lines for the marginal family.

2. *Self-similar versus self-affine boundaries.* Theorem 2 uses the fact that boundary similarities extend to isometries of AdS. Anisotropic (self-affine) scalings do not, so self-affine boundaries such as (4) are not covered beyond the linearised problem. In the linearised model the effective exponent $D_{\text{fit}} \neq D$ of Section 3.6 is due to mode truncation and to the non-homogeneity of $\sqrt{1 + |\partial f|^2}$, not to a renormalisation of D .
3. *Distance-zeta approximation.* Theorem 3 rests on the landing-density approximation (32), which is exact for planar entangling surfaces only. Corrections of the type $g(z)$ change residues but, by Theorem 2, not the location of poles on the critical line.
4. *Scheme dependence.* Log-periodic terms are meaningful with respect to a fixed geometric cutoff and constant rescalings of it. A log-periodic reparametrisation of the cutoff can absorb them, as noted in [24].
5. *Cosmic-brane minimality.* The q -thread interpretation of Section 7 assumes that the backreacted brane $\mathcal{C}_q$ is minimal, not merely extremal, on the static slice of the backreacted geometry. This holds for spherical and planar regions, where the brane is the bifurcation surface of a hyperbolic black hole [4, 3]. It is not established for fractal entangling surfaces.
6. *Shape of the singularity spectrum.* In Einstein gravity the shape of $f(\alpha)$ is fixed by x_q , and in $d = 2$ it is the Calabrese–Lefevre spectrum. The factorisation of Rényi entropies for fractal surfaces in Section 6 is a leading-order approximation that we have not proven.
7. *Origin of non-trivial cascade weights.* Non-trivial multifractal weights p_j require sparse operator spectra, which are not described by classical Einstein gravity. In a holographic theory they can arise only at $O(c^0)$, through bulk quantum corrections [49] or superpositions of fixed-area states [50, 51]. Multifractal participation and entanglement spectra are well documented in other quantum many-body settings [54]. We have derived p_j only within the factorised BCFT model. The ideal Fermi-gas structure of Gaussian entanglement spectra and Markov-modulated tensor cascades are analysed in a companion paper [55].
8. *Other observables.* Transport observables of fractal Josephson-junction arrays, such as critical currents as a function of flux, are governed by a different (decimation) structure and are not covered by our analysis.
9. *Computability of the dimension.* Theorems 1–3 take the dimension D_∂ as given. For self-similar boundaries with finitely many computable ratios it is computable, but a com-

putable, non-self-similar boundary can have an uncomputable dimension, while its discretisations remain algorithmically simple at every finite resolution [56].

Data and code availability

The Python script `holographic_fractal_rt.py` produces all numerical results and figures in this paper. It is run in the modes `rt`, `ren`, `renyi`, `stage3` and `stage4`, and it is distributed in the repository <https://github.com/Ruqing1963/holographic-fractal-entanglement> and archived on Zenodo under DOI [10.5281/zenodo.23189575](https://doi.org/10.5281/zenodo.23189575).

A Numerical methods

All computations use `holographic_fractal_rt.py` with NumPy 1.26, SciPy 1.13, SymPy 1.13 and Matplotlib 3.8. The script has five modes: `rt` (Section 3, Table 1), `ren` (Section 4.5), `renyi` (Section 6), `stage3` (Sections 5–7) and `stage4` (Sections 6 and 8).

- *Symbolic checks.* SymPy is used to compute the Christoffel symbols and the Riemann and Ricci tensors of (2) in $d = 4$, to verify that ϕ_3 solves (8), and to verify $\nabla_\mu v^\mu = 0$ for the geodesic flow (51) for general d .
- *Linearised surfaces and content.* The mode sums for f , $\partial_y f$ and f_z are evaluated directly on a uniform periodic grid of 2^{15} points per period (2^{16} in Section 4.5), with ϕ_d and ϕ'_d expressed through modified Bessel functions. $\mathcal{A}(z)$ is the Riemann sum over the grid, which is spectrally accurate for smooth periodic integrands. Radial integrals use the trapezoidal rule in $\ln z$.
- *Complex-dimension fits.* The expansion of Section 4.5 is fitted by linear least squares, with D scanned over 401 values in $[D - 0.2, D + 0.2]$ and the minimum residual retained. The counterterm (29) is then applied analytically to the fitted terms, using $(-z\partial_z)^{-1}z^{-\omega} = z^{-\omega}/\omega$.
- *Moran roots and the renewal model.* Roots of Moran polynomials are computed as companion-matrix eigenvalues (`numpy.roots`) and mapped to $s = -\ln x / \ln b + 2\pi im / \ln b$. In the exactly solvable renewal model of Section 5, $S(\varepsilon)$ is evaluated in closed form through exponential integrals E_1 and the word counts c_K .
- *Cascades and Rényi entropies.* Cascade spectra are enumerated exactly by composition, with logarithmic binomial coefficients. Traces are evaluated by log-sum-exp, and $\tau(q)$ is obtained by Brent’s method. The hyperbolic-brane radii x_q are also found by Brent’s method, and Dong’s identity is checked by central differences with step 10^{-6} .
- *BCFT spectra.* The Cardy density is discretised on 4×10^5 points.
- *Resonator.* The Weyl-subtracted heat trace is evaluated without cancellation using the Jacobi transformation of the theta function. Γ and the Riemann ζ at complex arguments are evaluated with `mpmath`.

B Mellin asymptotics and renewal theory

We record the standard argument [22, 15] behind Theorem 2(iv). Define

$$F(\ell) = e^{-(D_\partial - (d-2))\ell} \mathcal{A}(e^{-\ell}), \quad w_j = r_j^{D_\partial}, \quad \sum_j w_j = 1. \quad (61)$$

Then the renewal equation (24) reads

$$F(\ell) = \sum_j w_j F(\ell - \lambda_j) + \tilde{g}(\ell). \quad (62)$$

For the Laplace transform $\widehat{F}(s) = \int_0^\infty e^{-s\ell} F(\ell) d\ell$ this gives

$$\widehat{F}(s) = \frac{\widehat{g}(s) + E(s)}{1 - \sum_j w_j e^{-s\lambda_j}}, \quad (63)$$

with E entire. The denominator vanishes at $s = 0$. On the line $\operatorname{Re} s = 0$ its zeros are exactly $s \in i\Lambda^\perp$, by the convexity argument of Theorem 2, Step 3. In the lattice case $\lambda_j = k_j \ln b$, the denominator is periodic under $s \mapsto s + 2\pi i / \ln b$.

Suppose that $\widehat{g}$ is holomorphic for $\operatorname{Re} s > -\eta''$ and of polynomial growth on vertical lines there, and that the denominator has no further zeros in $-\eta'' < \operatorname{Re} s < 0$. Shifting the Bromwich contour of the inverse Laplace transform to $\operatorname{Re} s = -\eta''$ then gives

$$F(\ell) = \sum_{t \in \Lambda^\perp} \operatorname{Res}_{s=it} [\widehat{F}(s) e^{s\ell}] + O(e^{-\eta''\ell}), \quad (64)$$

where the sum is understood in the sense of distributions in ℓ or, under stronger decay of $\widehat{g}$, pointwise. This is the Fourier series of the periodic function $P(\ell)$, with frequencies in $\Lambda^\perp = (2\pi / \ln b)\mathbb{Z}$. Without the growth hypothesis, the lattice renewal theorem [22] still gives $F(\ell) - P(\ell) \rightarrow 0$, but without a rate. In the non-lattice case, $\Lambda^\perp = \{0\}$ and F converges to a constant.

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